

Enrique Sobrados

PhD Student, Computer Science, *University of New Mexico*

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RESEARCH OVERVIEW

My research explores measurement-driven security and privacy issues in networked systems and mobile communications. I build automated large-scale data collection pipelines, LLM-based security decisions, and study how configuration practices impact user security and privacy. Currently, I aim to implement AI Agents for large-scale detection of security misconfigurations.

EDUCATION

- PH.D., Computer Science, University of New Mexico, Albuquerque, NM, USA Present
Advisor: Dr. Afsah Anwar
- M.S., Computer Science, University of New Mexico, Albuquerque, NM, USA 2025
- B.S., Computer Science, University of Engineering and Technology, Lima, Peru 2023

PUBLICATIONS

1. Wang*, W., Ortwein*, A., **Sobrados***, E., Stanley, R., Kumar, P., Anwar, A., and Ensafi, R. "MVPNAlyzer: An Investigative Framework for the Security & Privacy Audit of Mobile VPNs". (*NDSS 2026*) (* joint first authors)
Built an investigative framework for large-scale security and privacy audits of mobile VPN ecosystems. Analyzed VPN app backends and infrastructure to identify systemic vulnerabilities, protocol exposures, and privacy leaks. Developed automated tools to measure the efficacy of VPN security claims across diverse network conditions.
2. **Sobrados**, E., Danish, M., Vanylssel, J., Ensafi, R., and Anwar, A. "Fingerprinting Probe-Resistant VPNs via Protocol Coexistence". (*Under Submission - S&P Oakland 2027*)
Reverse-engineered 20 VPN apps API backend to enumerate VPN servers, collecting 71,862 server IPs. Found that $\approx 80\%$ of servers are trivially detectable as VPN infrastructure, and that "probe-resistant" protocols frequently coexist on the same servers with legacy and fingerprintable protocols, undermining censorship-resistance claims.
3. Danish, M., **Sobrados**, E., Acharya, B., Saad, M., Mueen, A., Rahaman, S., and Anwar, A. "Private Links, Public Leaks: Consequences of Frictionless User Experience on the Security and Privacy Posture of SMS-Delivered URLs". (*Under Submission - CCS 2026*)
Analyzed 322K URLs from 33M SMS messages. Built an end-to-end pipeline (data collection, LLM-based PII detection, and manual validation) to identify 150+ PII-exposing services. Uncovered weak bearer-token authentication, low-entropy token user enumeration, privileged-action exposures, and account takeover. Responsible disclosures protected 150M+ users.
4. Vanylssel, J., **Sobrados**, E., Anwar, R., Roman, G.C., and Anwar, A. "Architectural Trust Collapse in Modular Space Flight Software". (*Under Submission - NDSS 2027*)
Developed a comprehensive threat model for the SmallSat supply chain, identifying critical vulnerabilities in COTS (Commercial Off-The-Shelf) hardware and open-source flight software. Proposed a hardware-rooted defense-in-depth framework to mitigate supply chain contamination and unauthorized command execution in NewSpace constellations.
5. **Sobrados**, E., Danish, M., Ensafi, R., and Anwar, A. "IP Contamination: How VPN servers contaminate the Internet". (*In Progress*)
Ongoing analysis of how churned VPN servers remain with low security reputation after they stopped being used as VPNs.

PROFESSIONAL EXPERIENCE

- **Graduate Research Assistant**, University of New Mexico Present
 - Built **MVPNAlyzer**: extensible mobile-VPN auditing framework (DNS leaks, obfuscation, cleartext traffic, and OpenVPN configurations).

- Disclosed vulnerabilities to **40+** VPN providers, with **6** acknowledging vulnerabilities so far.
- Enumerated infrastructure for **~60,000** VPN servers; reverse-engineered APIs to map **20+** endpoints.
- Uncovered **protocol weaknesses** present on VPN infrastructure with fingerprintable protocols.
- **Software Engineer (Project Consultant)**, Meta-Innovaciones S.A.C., Peru 2023
 - Developed the front-end for **multiple** intake desk processes, including administrative litigation.
 - Performed secure code reviews and remediated **4 security** weaknesses.
 - Automated taxpayer report generation using Jasper Reports + SpringBoot Backend, reducing manual processing effort by over **80%**.
 - Assisted in designing a **scalable database** architecture for multiple system modules, improving data consistency.
- **Security Analyst Intern**, National Bank, Peru 2022
 - Operated VPNs on **Palo Alto & Check Point** (IPsec/IKEv2) for **~2,000** users.
 - Tuned **Forcepoint** web filtering; cut false positives **15%** while preserving business access.
 - Automated ATM key generation/rotation in **Bash**; standardized ceremonies; saved **6 hrs/week**.
 - Ran vuln triage across **2** services; **5** issues remediated in **4** weeks.

TEACHING

- **Guest Lecturer**, University of New Mexico — Intro to Cybersecurity (CS544) 2025
- **Teaching Assistant**, University of New Mexico — Intro to Cybersecurity (CS544) 2024
- **Teaching Assistant**, University of New Mexico — Declarative Programming (CS357L) 2023
- **Teaching Assistant**, Le Wagon — Web Development Bootcamp 2021
- **Teaching Assistant**, University of Engineering and Technology — Intro to Object-Oriented Programming 2020

SERVICE

1. Co-Reviewer, The Network and Distributed System Security Symposium (NDSS) 2026
2. Co-Reviewer, Annual Computer Security Applications Conference (ACSAC) 2025

CERTIFICATIONS

1. Junior Penetration Tester (eJPTv2)
 2. PEN-100, SOC-100, WEB-100
- INE Security
Offensive Security

AWARDS & LEADERSHIP

- **President**, LoboSec (UNM CTF Team) Present
- **NDSS Fellowship**, Internet Society 2026
- **3rd Place**, Lobo Hackathon 2026
- **1st place — Top Scorer**, Alamo CTF 2025
- **CS Department Award**, University of New Mexico 2024
- **1st Place — Tracer Fire CTF**, Sandia National Laboratories 2023
- **ACSAC Student Conferenceship** 2023

REFERENCES

Afsah Anwar, Assistant Professor
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